



Calibrate crop growth

Estimate shared crop parameters
Evaluate development and growth



Calibrate crop water use

Fit irrigation-treatment responses
Retrospectively test crop water use

Simulate continuous regional rotations

Carry soil water across wheat, maize and fallows
Compare crop responses at five irrigation levels



Allocate irrigation spatially

Compare uniform and targeted allocation
Hold the regional water budget constant
Selection: 1997–2013

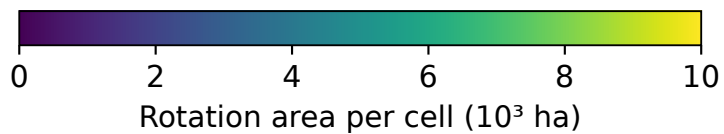
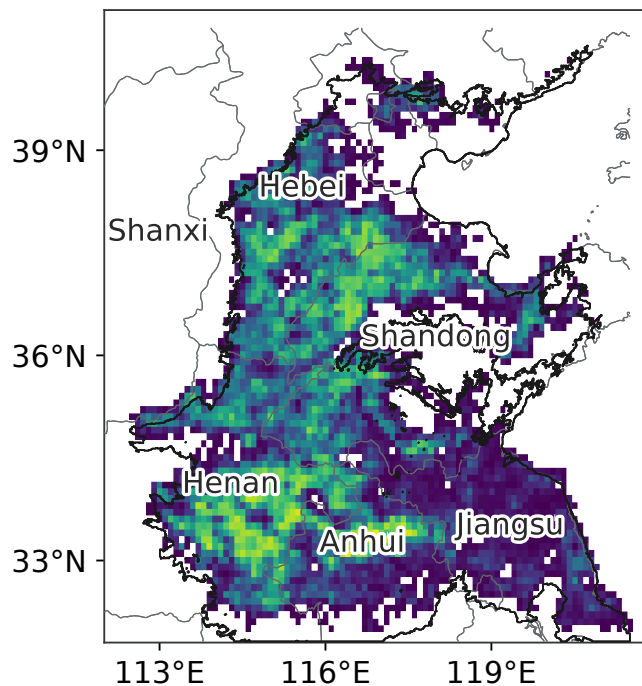
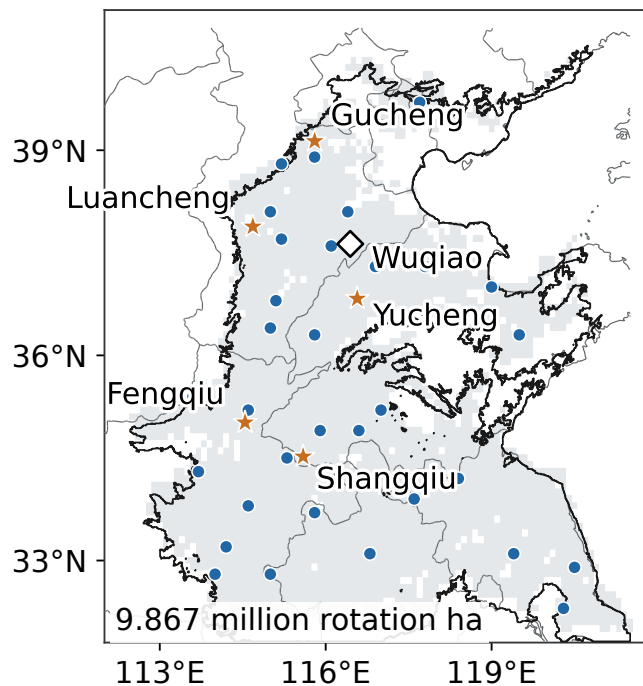
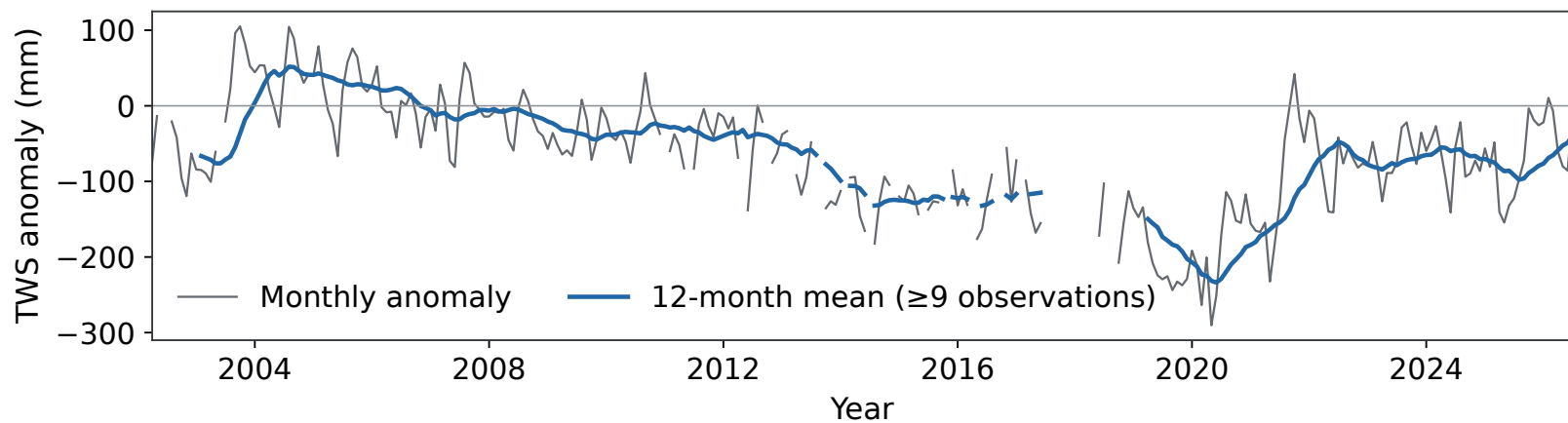


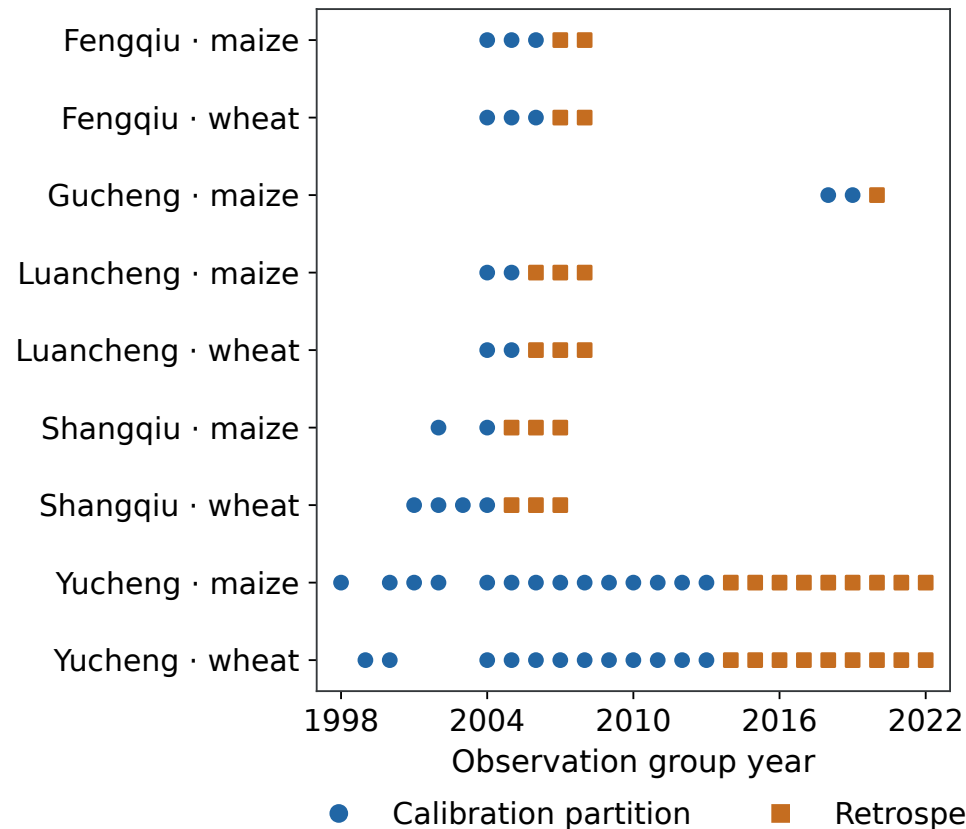
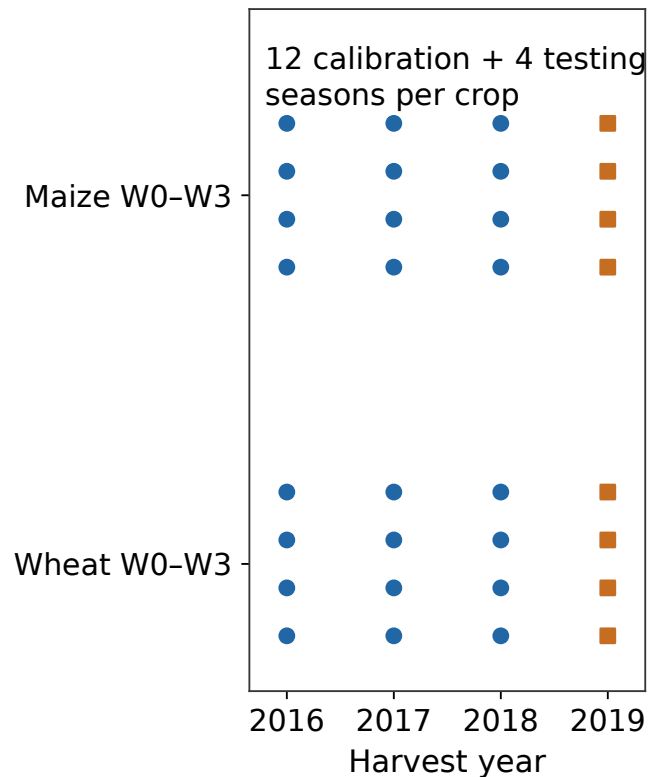
Select annual quotas

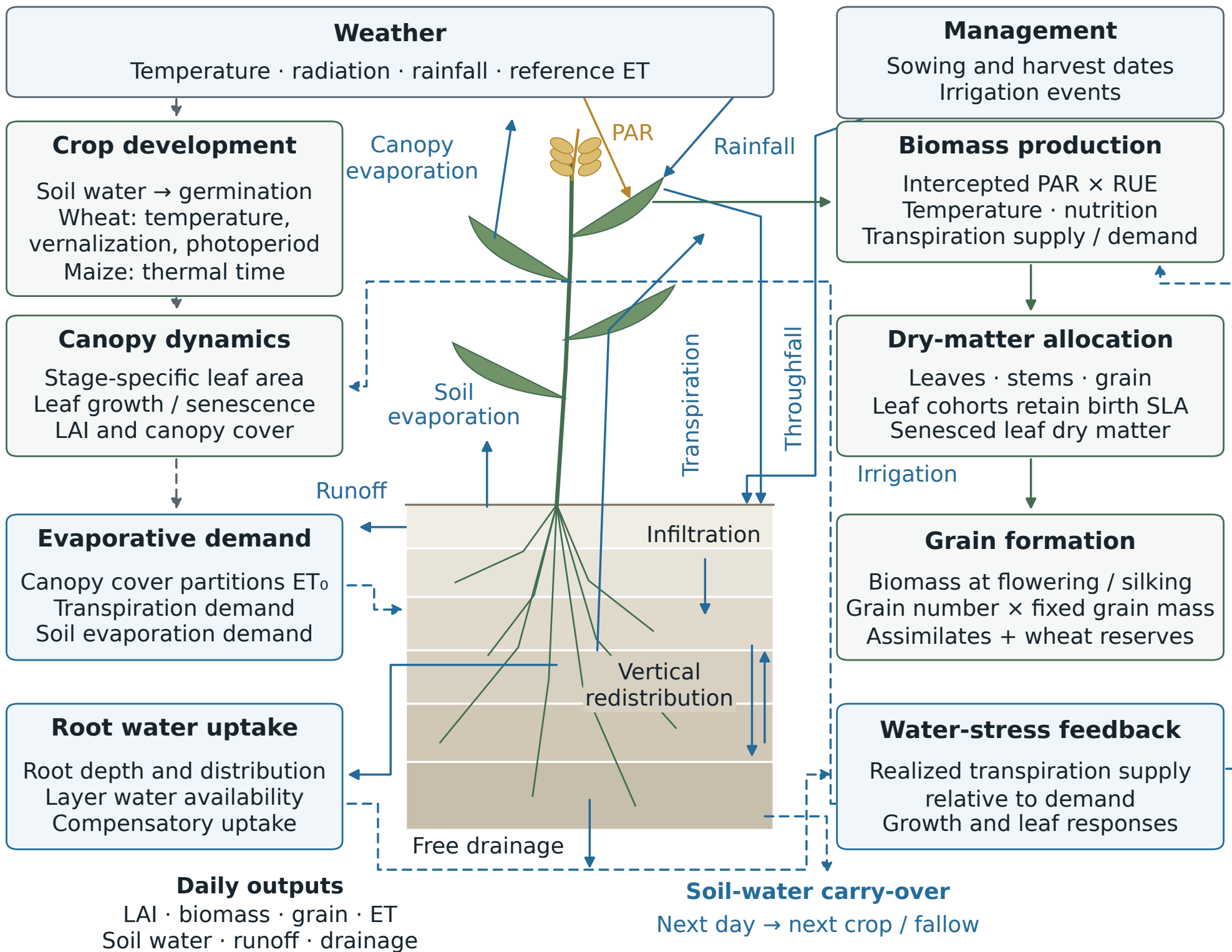
Compare storage-rainfall and rainfall-only
Match monitoring availability
Selection: 2003–2013

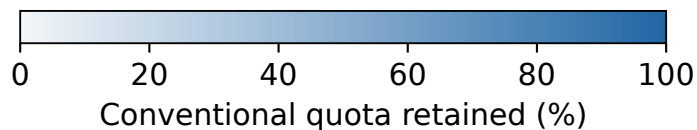
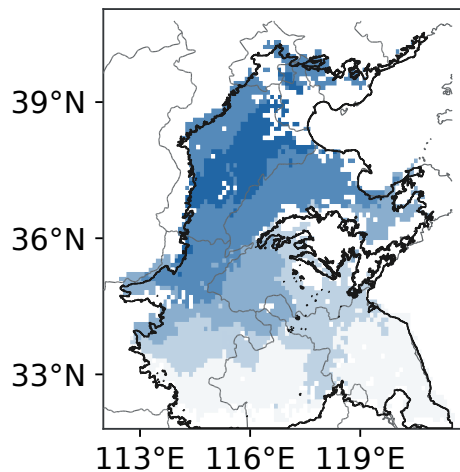
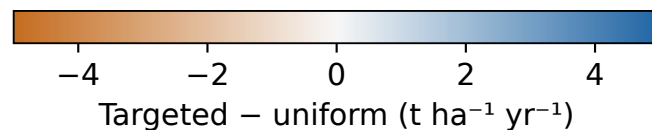
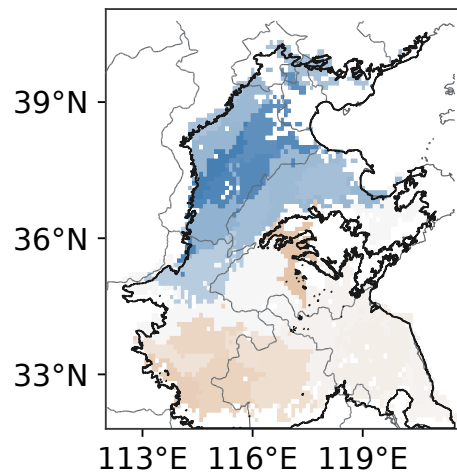
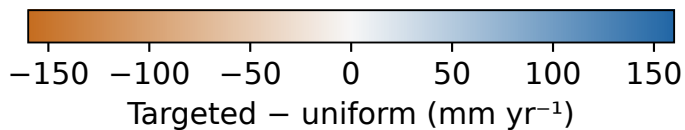
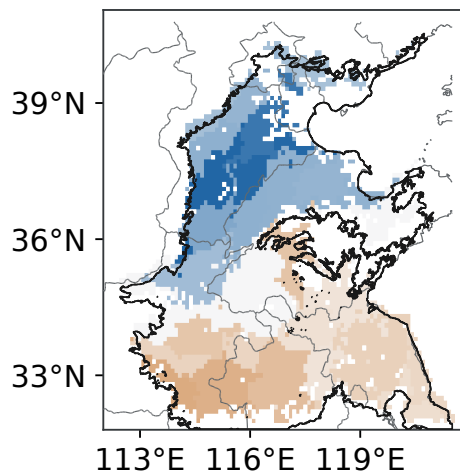
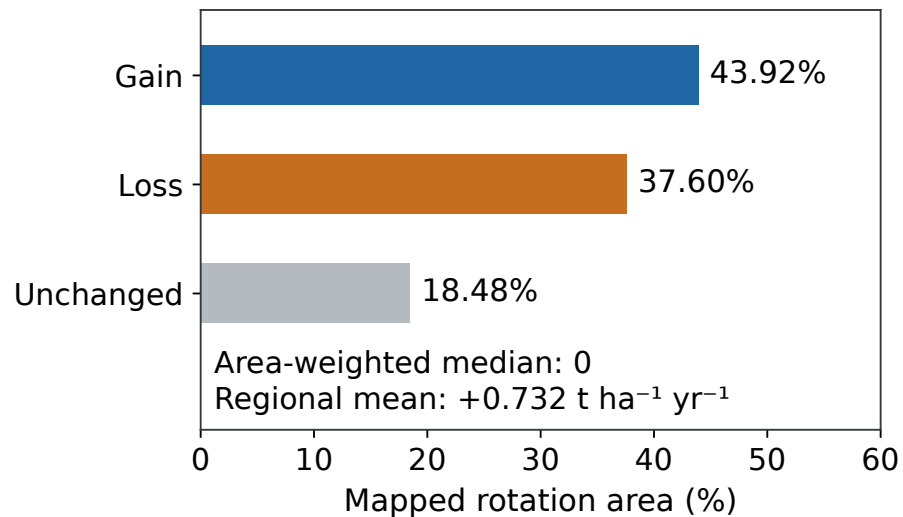
Evaluate management outcomes: 2014–2025

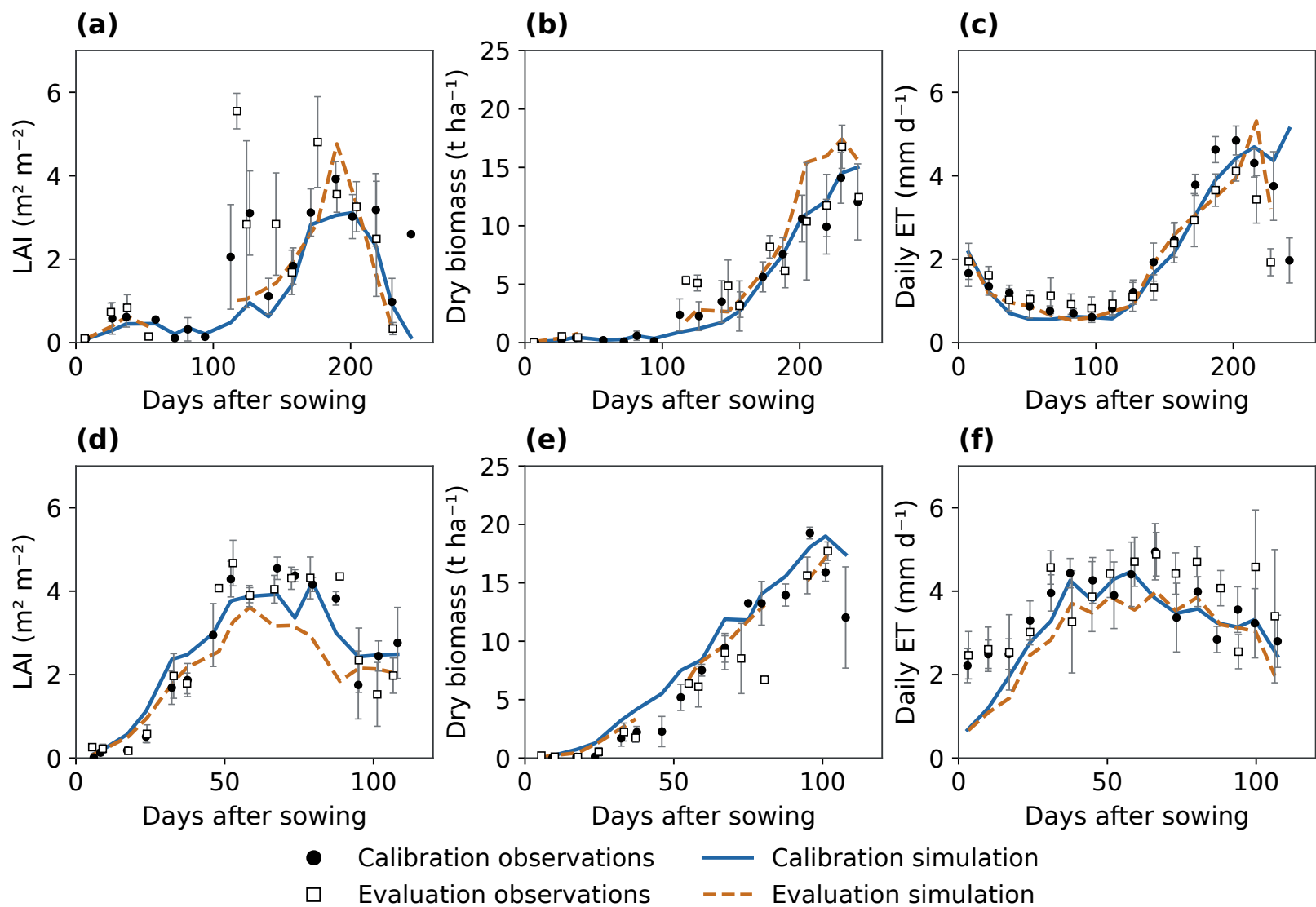
Rotation grain · irrigation · ET · drainage · spatial gains and losses

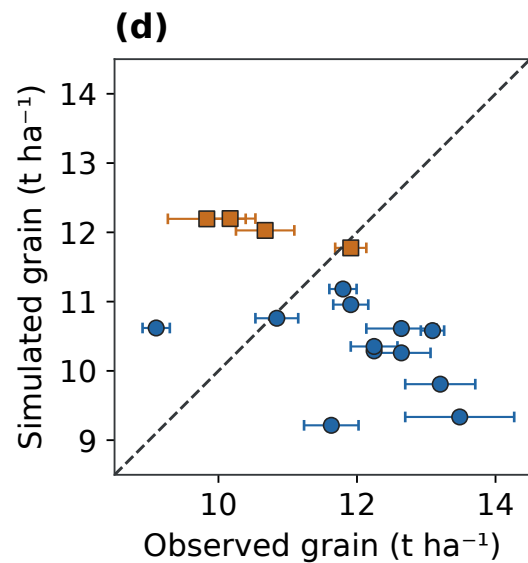
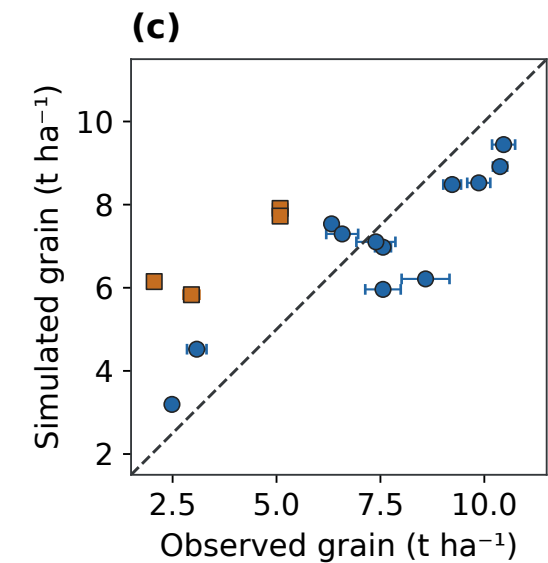
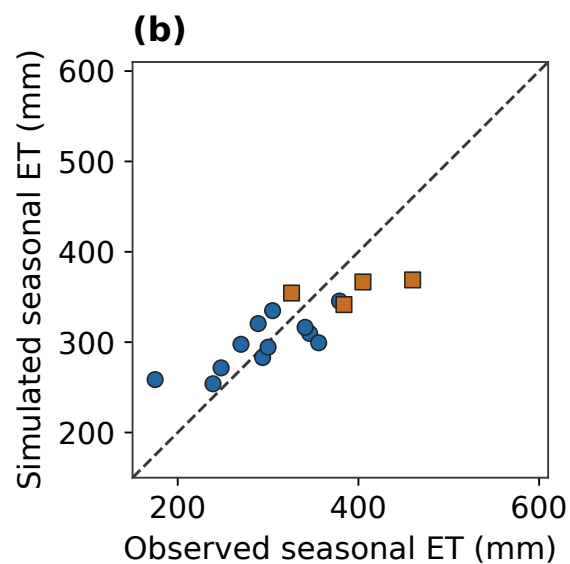
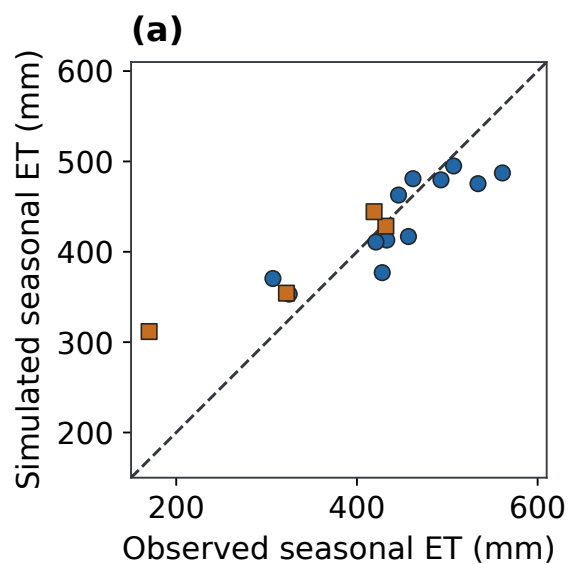
(a)**(b)****(c)**

(a)**(b)**

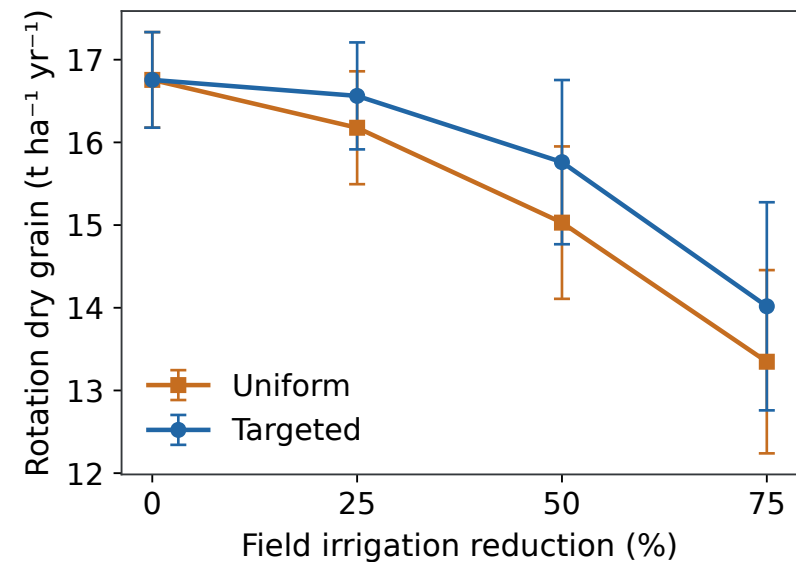
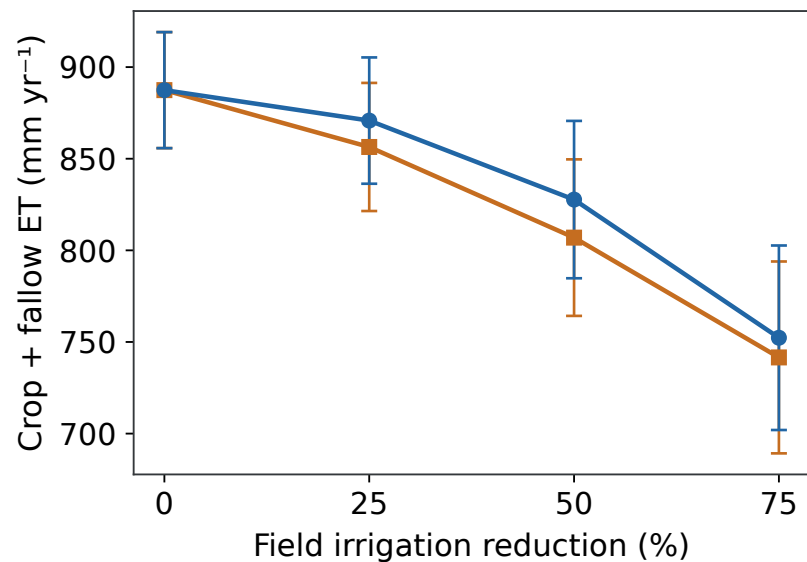
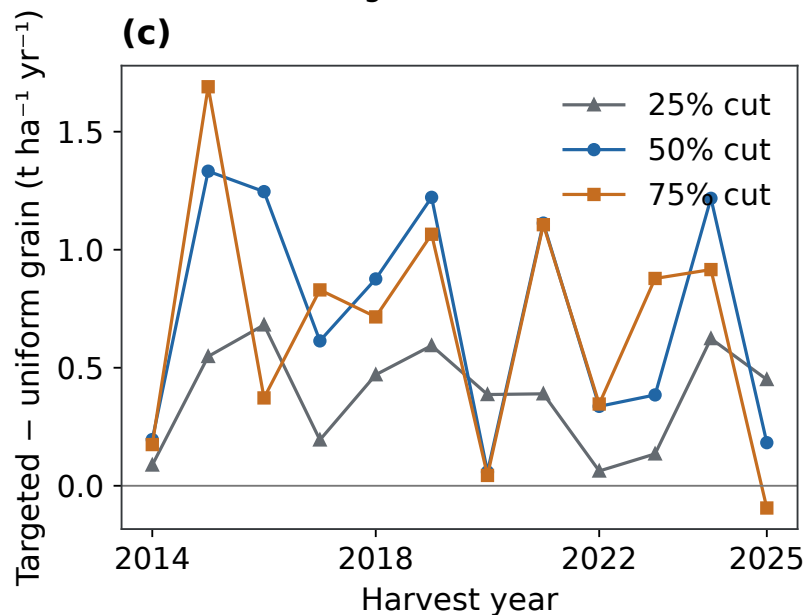


(a)**(b)****(c)****(d)**





- Calibration: 2016-2018 ($n = 12$ per crop)
- Retrospective testing: 2019 ($n = 4$ per crop)

(a)**(b)****(c)****(d)**